Measure Theoretic Probability
Lesson 23
Expectation of Simple Random Variables Is Well Defined

Kirschner Awonatey Amaanya

Department of Statistics and Actuarial Science, KNUST

Outline

1. The Question: Does the Representation Matter?
2. Examples
3. The Canonical Partition and Locked-In Conditions
4. Expectation Is Well Defined
5. First Properties of Expectation
6. Summary and Exercises

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Recap and Question

Simple random variable and its expectation

$(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space. A simple random variable has the form

$$X(\omega) = \sum_{i=1}^{n} x_i \, I_{A_i}(\omega), \qquad x_i \in \mathbb{R}, \; A_i \in \mathcal{F}, \qquad \mathbb{E}[X] := \sum_{i=1}^{n} x_i \, \mathbb{P}(A_i).$$

Worry

The same function X can be written in many ways. If you use one representation and I use another, do we get the same $\mathbb{E}[X]$?

Plan

Look at examples first, then prove that $\mathbb{E}[X]$ does **not** depend on the representation.

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Example 1: Splitting a Set into Two Pieces

Setup

$\Omega = [0,1]$ (the σ-field and $\mathbb{P}$ do not matter here; think Borel sets and Lebesgue measure).

$$X = 2\, I_{[0,\frac{1}{3})} + 3\, I_{[\frac{1}{3},\frac{2}{3})} + 4\, I_{[\frac{2}{3},1]}.$$

Another representation

Split $[0,\frac{1}{3}) = [0,\frac{1}{4}) \cup [\frac{1}{4},\frac{1}{3})$:

$$X = 2\, I_{[0,\frac{1}{4})} + 2\, I_{[\frac{1}{4},\frac{1}{3})} + 3\, I_{[\frac{1}{3},\frac{2}{3})} + 4\, I_{[\frac{2}{3},1]}.$$

Same function: every ω goes to the same value $2, 3$ or 4.

Expectations agree

$$2\mathbb{P}[0,\tfrac{1}{4}) + 2\mathbb{P}[\tfrac{1}{4},\tfrac{1}{3}) = 2\big(\mathbb{P}[0,\tfrac{1}{4}) + \mathbb{P}[\tfrac{1}{4},\tfrac{1}{3})\big) = 2\,\mathbb{P}[0,\tfrac{1}{3}),$$

by **finite additivity**, since the two pieces are disjoint. The other terms are unchanged.

Example 2: A Part of Ω Seems Missing

Representation with two terms

$$X = 2\, I_{[0,\frac{1}{3})} + 3\, I_{[\frac{1}{3},\frac{2}{3})} \qquad \text{on } \Omega = [0,1].$$

The sets do not cover $[\frac{2}{3}, 1]$. But for $\omega \in [\frac{2}{3}, 1]$ both indicators are 0, so $X(\omega) = 0$: X is defined on all of Ω.

Add the missing piece with coefficient 0

$$X = 2\, I_{[0,\frac{1}{3})} + 3\, I_{[\frac{1}{3},\frac{2}{3})} + 0 \cdot I_{[\frac{2}{3},1]}.$$

Now the three sets partition Ω. The expectation formula gives the same value, since the new term is $0 \cdot \mathbb{P}[\frac{2}{3}, 1] = 0$.

Remark

Last time we said the sets A_i should cover Ω. With indicator notation this is automatic, because any uncovered part carries the value 0 and can be added as an extra set.

Example 3: Overlapping Sets

A representation with overlaps

$$X = 2\,I_{[0,\frac{1}{3})} + 3\,I_{[\frac{1}{4},\frac{2}{3})} + 4\,I_{[\frac{2}{3},1]}.$$

The first two intervals overlap on $[\frac{1}{4}, \frac{1}{3})$. For $\omega = 0.26$ both indicators are on, so $X(\omega) = 2 + 3 = 5$. So X takes the value 5, which is not a coefficient.

Canonical form (distinct values, true partition)

Set	Value	Length
$[0, \frac{1}{4})$	2	$\frac{1}{4}$
$[\frac{1}{4}, \frac{1}{3})$	5	$\frac{1}{12}$
$[\frac{1}{3}, \frac{2}{3})$	3	$\frac{1}{3}$
$[\frac{2}{3}, 1]$	4	$\frac{1}{3}$

Both computations (Lebesgue measure)

$$2 \cdot \tfrac{1}{3} + 3 \cdot \tfrac{5}{12} + 4 \cdot \tfrac{1}{3} = \tfrac{39}{12} \quad \text{and} \quad 2 \cdot \tfrac{1}{4} + 5 \cdot \tfrac{1}{12} + 3 \cdot \tfrac{1}{3} + 4 \cdot \tfrac{1}{3} = \tfrac{39}{12}.$$

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The Canonical Partition

Definition (terminology of this course)

$X = \sum_{i=1}^{n} x_i I_{A_i}$ is in **canonical form** if

- the values $x_1, \ldots, x_n$ are **distinct**, and
- $A_i = X^{-1}(\{x_i\}) = \{X = x_i\}$, so $A_1, \ldots, A_n$ partition Ω.

Any simple random variable can be put in this form: merge terms with equal coefficients, and split overlaps into their own sets.

It matches the familiar formula

Since $\mathbb{P}(A_i) = \mathbb{P}(X = x_i)$ (shorthand for $\mathbb{P}(\{\omega : X(\omega) = x_i\})$),

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \, \mathbb{P}(A_i) = \sum_{i=1}^{n} x_i \, \mathbb{P}(X = x_i).$$

Remark

"Canonical partition" is this course's own phrase, not standard terminology in textbooks.

Two Representations Lock in Conditions

Setup

Let $X = \sum_{i=1}^{n} x_i I_{A_i} = \sum_{j=1}^{m} y_j I_{B_j}$, with $\{A_i\}$ canonical and $\{B_j\}$ another partition of Ω.

Key observation

If $\omega \in A_i \cap B_j$, then the first representation gives $X(\omega) = x_i$ and the second gives $X(\omega) = y_j$. Hence

$$A_i \cap B_j \neq \varnothing \implies x_i = y_j.$$

Consequence

Suppose some B_1 met both A_1 and A_2. Then $x_1 = y_1 = x_2$, contradicting the distinctness of the canonical coefficients. So every B_j lies inside a single A_i: **any other partition representing X is a subpartition of the canonical one.**

Subpartitions Give the Same Expectation

Claim

If $\{B_j\}$ refines the canonical partition $\{A_i\}$, the two expectations agree.

Proof

Say B_1, B_2, B_3 are the pieces of A_1. They all map to x_1, so $y_1 = y_2 = y_3 = x_1$. Then

$$y_1\mathbb{P}(B_1) + y_2\mathbb{P}(B_2) + y_3\mathbb{P}(B_3) = x_1\big(\mathbb{P}(B_1) + \mathbb{P}(B_2) + \mathbb{P}(B_3)\big) = x_1\,\mathbb{P}(A_1),$$

by finite additivity, since $A_1 = B_1 \cup B_2 \cup B_3$ is a disjoint union. Doing this for each A_i gives equality of the two sums. □

Remark

Sets in a representation with $\mathbb{P}(B_j) = 0$ contribute 0 and are harmless, even if they are nonempty.

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Theorem: Independence of the Partition

Theorem

Let $X = \sum_{i=1}^{n} x_i I_{B_i} = \sum_{j=1}^{m} y_j I_{C_j}$, where $\{B_i\}$ and $\{C_j\}$ are two partitions of Ω into sets of $\mathcal{F}$ (neither need be canonical, and neither need refine the other). Then

$$\sum_{i=1}^{n} x_i \, \mathbb{P}(B_i) = \sum_{j=1}^{m} y_j \, \mathbb{P}(C_j).$$

Ingredient

As before: $B_i \cap C_j \neq \varnothing \Rightarrow x_i = y_j$.

Proof of the Theorem

Step 1: cut each set along the other partition

$\{C_j\}$ partitions Ω, so $B_i = \bigsqcup_{j=1}^{m}(B_i \cap C_j)$, a disjoint union (some pieces may be empty). By finite additivity,

$$\sum_{i=1}^{n} x_i \mathbb{P}(B_i) = \sum_{i=1}^{n}\sum_{j=1}^{m} x_i \, \mathbb{P}(B_i \cap C_j).$$

Likewise, since $\{B_i\}$ partitions Ω,

$$\sum_{j=1}^{m} y_j \mathbb{P}(C_j) = \sum_{j=1}^{m}\sum_{i=1}^{n} y_j \, \mathbb{P}(B_i \cap C_j) = \sum_{i=1}^{n}\sum_{j=1}^{m} y_j \, \mathbb{P}(B_i \cap C_j).$$

(The order of finite sums may be exchanged.)

Step 2: compare the (i, j) terms

- $B_i \cap C_j = \varnothing$: both terms are 0.
- $B_i \cap C_j \neq \varnothing$: then $x_i = y_j$, so $x_i \mathbb{P}(B_i \cap C_j) = y_j \mathbb{P}(B_i \cap C_j)$ (even if the probability is 0).

All nm terms agree, so the two sums are equal. □

Overlapping Sets Are Also Fine

Representation with overlaps

Let $X = \sum_{j=1}^{m} y_j I_{B_j}$ with arbitrary sets $B_j \in \mathcal{F}$ (e.g. $m = 3$, drawn as a Venn diagram).

Pass to atoms

For $S \subseteq \{1, \dots, m\}$ let $D_S = \bigcap_{j \in S} B_j \cap \bigcap_{j \notin S} B_j^c$. These sets are disjoint, lie in $\mathcal{F}$, and cover Ω. On D_S we have $X = \sum_{j \in S} y_j$. Then

$$\sum_S \Big(\sum_{j \in S} y_j \Big) \mathbb{P}(D_S) = \sum_{j=1}^{m} y_j \sum_{S \ni j} \mathbb{P}(D_S) = \sum_{j=1}^{m} y_j \, \mathbb{P}(B_j),$$

because $B_j = \bigsqcup_{S \ni j} D_S$ (finite additivity again).

Conclusion

The "overlap" representation and the atom representation give the same sum. Merging atoms with equal values then gives the canonical form, by the theorem. So $\mathbb{E}[X]$ is well defined for *every* representation.

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Setting for the Properties

Convention

Throughout, X is a **simple** random variable in canonical form: $X = \sum_{i=1}^{n} x_i I_{A_i}$, with x_i distinct and $A_i = \{X = x_i\}$ partitioning Ω.

Meaning of "almost surely"

$X = 0$ a.s. means $\mathbb{P}(\{\omega : X(\omega) = 0\}) = 1$. X may take other values, but only on sets of probability 0. Similarly, $X \geq 0$ a.s. means $\mathbb{P}(X \geq 0) = 1$.

Remark

We prove these only for simple random variables, since $\mathbb{E}$ is not yet defined for general ones. They will hold for general random variables.

Property 1: $X = 0$ a.s. $\Rightarrow \mathbb{E}[X] = 0$

Proof

$\mathbb{P}(X = 0) = 1 > 0$, so 0 is one of the values; say $x_1 = 0$ and $A_1 = \{X = 0\}$. Then $\mathbb{P}(A_1) = 1$, and since the A_i partition Ω,

$$\sum_{i \geq 2} \mathbb{P}(A_i) = 1 - \mathbb{P}(A_1) = 0, \quad \text{so} \quad \mathbb{P}(A_i) = 0 \ \ (i \geq 2).$$

Therefore

$$\mathbb{E}[X] = 0 \cdot \mathbb{P}(A_1) + \sum_{i=2}^{n} x_i \cdot 0 = 0. \qquad \square$$

Remark

The values x_i with $i \geq 2$ can be anything (e.g. 4). They sit on sets of probability 0, so they do not contribute.

Property 2: $X \geq 0$ a.s. $\Rightarrow \mathbb{E}[X] \geq 0$

Proof

If $x_i < 0$, then $A_i = \{X = x_i\} \subseteq \{X < 0\}$, which has probability 0, so $\mathbb{P}(A_i) = 0$. Hence in

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \, \mathbb{P}(A_i)$$

every term is either $x_i \mathbb{P}(A_i) \geq 0$ (when $x_i \geq 0$) or 0 (when $x_i < 0$). So $\mathbb{E}[X] \geq 0$. □

Idea

Negative values can occur, but only on null sets, and null sets contribute nothing to the sum.

Property 3: $X \geq 0$ a.s. and $\mathbb{E}[X] = 0 \Rightarrow X = 0$ a.s.

Proof

As in Property 2, every term $x_i \mathbb{P}(A_i)$ is ≥ 0 (terms with $x_i < 0$ are 0). A sum of nonnegative terms equal to 0 forces every term to be 0:

$$x_i \, \mathbb{P}(A_i) = 0 \quad \text{for each } i,$$

so for each i, either $x_i = 0$ or $\mathbb{P}(A_i) = 0$. Therefore

$$\mathbb{P}(X \neq 0) = \sum_{i:\, x_i \neq 0} \mathbb{P}(A_i) = 0,$$

i.e. $X = 0$ almost surely. □

Nonnegativity is essential

A random variable can have $\mathbb{E}[X] = 0$ without being 0, e.g. $X = \pm 1$ with probability $\frac{1}{2}$ each: positive and negative terms cancel.

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Summary

Key results

Concept / Result	Content
Definition	$X = \sum x_i I_{A_i}$, $\mathbb{E}[X] = \sum x_i \mathbb{P}(A_i)$
Canonical form	x_i distinct, $A_i = \{X = x_i\}$ partition Ω; $\mathbb{E}[X] = \sum x_i \mathbb{P}(X = x_i)$
Locked-in condition	$A_i \cap B_j \neq \varnothing \Rightarrow x_i = y_j$
Subpartitions	any other partition refines the canonical one; same $\mathbb{E}[X]$ by finite additivity
Well-definedness	two arbitrary partitions: compare the nm terms $\mathbb{P}(B_i \cap C_j)$
Overlaps	pass to atoms D_S; same sum
Property 1	$X = 0$ a.s. $\Rightarrow \mathbb{E}[X] = 0$
Property 2	$X \geq 0$ a.s. $\Rightarrow \mathbb{E}[X] \geq 0$
Property 3	$X \geq 0$ a.s., $\mathbb{E}[X] = 0 \Rightarrow X = 0$ a.s.

Exercises

1. Verify that $\frac{39}{12}$ is the expectation in Example 3 using both representations.
2. Let $X = I_A + I_B$ with $A, B \in \mathcal{F}$. Write X in canonical form and compute $\mathbb{E}[X]$ in terms of $\mathbb{P}(A), \mathbb{P}(B), \mathbb{P}(A \cap B)$.
3. Show that $\mathbb{E}[I_A] = \mathbb{P}(A)$ for every $A \in \mathcal{F}$.
4. Show that $\mathbb{E}[c] = c$ for a constant random variable $X \equiv c$.
5. In the proof of the main theorem, why may the order of the double sum be exchanged?
6. In the atom argument, show that the sets D_S are disjoint and that $B_j = \bigsqcup_{S \ni j} D_S$.
7. Give an example of a simple X with $\mathbb{E}[X] = 0$ but $\mathbb{P}(X = 0) < 1$.
8. In Property 3, where do you use that $X \geq 0$ almost surely?

Outlook

Next lesson

The remaining properties of expectation for simple random variables.

Looking ahead

We then return briefly to convergence, to limits of expectations, and afterwards extend expectation from simple random variables to **general** random variables.